Assessment Menu: Teacher Guide

The Logarithm Bistro · design notes, grading and answer key

Why students cannot just pick the easy ones

1. **Points are capped by category.** Students must complete exactly 4 / 4 / 2 dishes, so they cannot fill the total with Appetizers.
2. **Chef's Specials carry 40% of the grade** and Entrées another 40%. A student who only masters the basics tops out at 20 points.
3. **Skill coverage rule.** Their 10 picks must include at least one dish from each skill group below, so every strand of the unit is assessed.
4. **Harder dishes reward process.** Each has a rubric that gives credit for correct steps, so a partly correct attempt at a harder problem earns more than a perfect easy one in the same time.

Skill group	What it covers	Dishes
EXPFUNC	Exponential functions, graphs and models	A1, A6, E1, S1
LOGDEF	Logarithm definition and evaluating	A2, A3, E2, S2
LOGPROP	Properties of logarithms	A4, E3, E4, S3
SOLVE	Solving exponential and logarithmic equations	A5, E5, E6, S4

If a student only does…	Maximum
Only Appetizers (4 × 5)	20
Appetizers + Entrées (4 × 5 + 4 × 10)	60
Full meal (adds 2 Chef's Specials)	100

Grading and options

- Grade only the marked dishes using the rubrics on the next pages (partial credit is built in). Total possible: **100**. If a student marks more than the required number, grade the first in each category.
- Check the skill rule on the order form before totaling. If a skill group is missing, apply your own policy (for example, the dish that broke the rule scores 0, or the student may swap one dish).
- Optional "second helping": allow one extra dish per category that replaces the student's lowest score in that category.
- Look at *which* dishes students chose. Students who select only the easiest dish in each skill group may need a conference.

Student	Appetizers (of 20)	Entrées (of 40)	Specials (of 40)	Skills OK?	Total

Answer Key: Appetizers (5 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
A1 EXPFUNC	For $y = 200(0.85)^x$: **(a)** Is this growth or decay? **(b)** What is the decay factor? **(c)** By what percent does y change each time x increases by 1?	**Answer:** (a) Decay (b) 0.85 (c) y decreases by 15% each step (a) The base 0.85 is less than 1, so this is **decay**. (b) The decay factor is the base, **0.85**. (c) $1 - 0.85 = 0.15$, so y **decreases by 15%** each time x increases by 1.	1 pts: (a) Decay, because the base is less than 1 2 pts: (b) Factor 0.85 (not 15% or 0.15) 2 pts: (c) 15% decrease ($1 - 0.85$), not 85%
A2 LOGDEF	Rewrite in the other form: **(a)** $5^3 = 125$ as a logarithm **(b)** $\log_7 49 = 2$ as an exponential equation	**Answer:** (a) $\log_5 125 = 3$ (b) $7^2 = 49$ (a) The base 5 stays the base of the log and the exponent 3 is the answer: $\log_5 125 = 3$ (b) The base 7 raised to the log value 2 gives 49: $7^2 = 49$	1 pts: (a) Base 5 written as the subscript 2 pts: (a) $\log_5 125 = 3$ 2 pts: (b) $7^2 = 49$
A3 LOGDEF	Evaluate without a calculator: **(a)** $\log_2 32$ **(b)** $\log_3 \frac{1}{27}$ **(c)** $\log_6 1$	**Answer:** (a) 5 (b) -3 (c) 0 (a) $2^5 = 32$, so $\log_2 32 = 5$ (b) $3^{-3} = \frac{1}{27}$, so the log is -3 (c) $6^0 = 1$, so $\log_6 1 = 0$	2 pts: (a) 5 2 pts: (b) -3 (negative exponent) 1 pts: (c) 0
A4 LOGPROP	Write as a single logarithm and simplify if possible: **(a)** $\log_2 6 + \log_2 5$ **(b)** $\log_5 40 - \log_5 8$	**Answer:** (a) $\log_2 30$ (b) $\log_5 5 = 1$ (a) Adding logs multiplies the inputs: $\log_2(6 \cdot 5) = \log_2 30$ (b) Subtracting logs divides the inputs: $\log_5 \frac{40}{8} = \log_5 5 = 1$	2 pts: (a) Multiplies 6 and 5, keeps base 2 2 pts: (b) Divides 40 by 8 1 pts: (b) Simplifies to 1
A5 SOLVE	Solve for x: **(a)** $5^x = 125$ **(b)** $2^{x+1} = 64$	**Answer:** (a) $x = 3$ (b) $x = 5$ (a) $125 = 5^3$, so $x = 3$ (b) $64 = 2^6$, so $x + 1 = 6$ and $x = 5$	2 pts: (a) Rewrites 125 as 5^3 and gets $x = 3$ 2 pts: (b) Rewrites 64 as 2^6 and sets $x + 1 = 6$ 1 pts: (b) $x = 5$ (not 6)
A6 EXPFUNC	For $y = 2^x - 3$: **(a)** Give the equation of the horizontal asymptote. **(b)** Find the y-intercept. **(c)** Is the function increasing or decreasing?	**Answer:** (a) $y = -3$ (b) $(0, -2)$ (c) increasing (a) 2^x gets close to 0 as $x \to -\infty$, so y gets close to -3: **$y = -3$** (b) $x = 0$: $2^0 - 3 = 1 - 3 = -2$, so **$(0, -2)$** (c) The base 2 is greater than 1, so the function is **increasing**.	2 pts: (a) $y = -3$ (the shift, not $y = 0$) 2 pts: (b) $(0, -2)$: substitutes $x = 0$ and evaluates $2^0 = 1$ 1 pts: (c) Increasing

Answer Key: Entrées (10 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
E1 EXPFUNC	$2,500 is invested at 6% annual interest. **(a)** Write A(t) if interest is compounded quarterly. **(b)** Find the balance after 5 years. **(c)** Find the balance after 5 years with continuous compounding, $A = Pe^{rt}$, and the difference. Round money to the nearest cent.	**Answer:** (a) $A(t) = 2500(1 + \frac{0.06}{4})^{4t}$ (b) $3367.14 (c) $3374.65, which is $7.51 more (a) r = 0.06 and n = 4, so $A(t) = 2500(1 + \frac{0.06}{4})^{4t} = 2500(1.015)^{4t}$ (b) $2500(1.015)^{20} \approx 3367.14$ (round only at the end) (c) $2500e^{0.3} \approx 3374.65$; the difference is about $7.51	3 pts: (a) Uses 0.06 ÷ 4 inside (not 1.06) and exponent 4t 3 pts: (b) Exponent 20 and $3367.14 2 pts: (c) $2500e^{0.3} = 3374.65$ 2 pts: (c) Difference $7.51
E2 LOGDEF	**(a)** Evaluate $\log_4 8$ without a calculator. **(b)** Solve $\log_x 81 = 4$, with $x > 0$. **(c)** Solve $\log_5 x = -2$. **(d)** Explain why $\log_3(-9)$ is undefined.	**Answer:** (a) $\frac{3}{2}$ (b) x = 3 (c) x = $\frac{1}{25}$ (d) no power of 3 equals −9 (a) $4^y = 8$ gives $2^{2y} = 2^3$, so $y = \frac{3}{2}$ (b) $x^4 = 81$, so x = 3 (a base is positive, so not −3) (c) $x = 5^{-2} = \frac{1}{25}$ (d) 3^y is always positive, so it never equals −9.	3 pts: (a) Rewrites with base 2 and gets $\frac{3}{2}$ 2 pts: (b) $x^4 = 81$ and x = 3 2 pts: (c) $5^{-2} = \frac{1}{25}$ 3 pts: (d) Says 3^y is always positive, so no y works
E3 LOGPROP	**(a)** Expand completely: $\log_3 \frac{9x^4}{y^2}$ **(b)** Write as one logarithm and simplify: 3 ln 2 + ln 5 − ln 10	**Answer:** (a) $2 + 4\log_3 x - 2\log_3 y$ (b) ln 4 (a) $\log_3 9 + \log_3 x^4 - \log_3 y^2 = 2 + 4\log_3 x - 2\log_3 y$ (b) $\ln 8 + \ln 5 - \ln 10 = \ln \frac{8 \cdot 5}{10} = \ln 4$	2 pts: (a) Evaluates $\log_3 9 = 2$ 2 pts: (a) $4\log_3 x$ (power rule) 2 pts: (a) $-2\log_3 y$ 2 pts: (b) Power rule: 3 ln 2 = ln 8 2 pts: (b) ln 4
E4 LOGPROP	Maya wrote $\log(x + 5) = \log x + \log 5$. **(a)** Explain her error and give a numeric counterexample. **(b)** Use change of base to find $\log_3 20$ to the nearest thousandth. **(c)** Between which two integers is $\log_3 20$? Justify.	**Answer:** (a) there is no rule for the log of a sum (b) ≈ 2.727 (c) between 2 and 3 (a) No rule splits a log of a sum. Example: $\log_2(4 + 4) = 3$, but $\log_2 4 + \log_2 4 = 4$. (b) $\log_3 20 = \frac{\ln 20}{\ln 3} \approx 2.727$ (c) $3^2 = 9 < 20 < 27 = 3^3$, so the log is between 2 and 3.	3 pts: (a) No rule for a sum, plus a valid counterexample 2 pts: (b) Setup $\frac{\ln 20}{\ln 3}$ 2 pts: (b) 2.727 3 pts: (c) Compares 20 with 3^2 and 3^3: between 2 and 3
E5 SOLVE	Solve using logarithms. Give the exact answer and then round to the nearest hundredth. **(a)** $7^x = 50$ **(b)** $4e^{2x} = 100$	**Answer:** (a) $x = \frac{\ln 50}{\ln 7} \approx 2.01$ (b) x = ln 5 ≈ 1.61 (a) $x \ln 7 = \ln 50$, so $x = \frac{\ln 50}{\ln 7} \approx 2.01$ (b) $e^{2x} = 25$, so $x = \frac{\ln 25}{2} = \ln 5 \approx 1.61$	3 pts: (a) Takes a log of both sides; exact $\frac{\ln 50}{\ln 7}$ 2 pts: (a) 2.01 3 pts: (b) Divides by 4 first, then ln; exact ln 5 2 pts: (b) 1.61
E6 SOLVE	Solve and check for extraneous solutions: **(a)**	**Answer:** (a) x = 4 (reject x = −2) (b) x = 5	2 pts: (a) One log: $x(x - 2) = 2^3 = 8$

	$\log_2 x + \log_2(x-2) = 3$ **(b)** $\log_3(2x-1) = 2$	(a) $x(x-2) = 2^3 = 8$, so $x^2 - 2x - 8 = 0$ and $x = 4$ or -2. $\log_2(-2)$ is undefined, so reject it: **$x = 4$**. (b) $2x - 1 = 9$, so $x = 5$. Check: $2(5) - 1 = 9 > 0$, valid.	2 pts: (a) Solves to $x = 4$ or -2 3 pts: (a) Rejects -2: log of a negative is undefined 3 pts: (b) $2x - 1 = 9$, $x = 5$, and checks it

Answer Key: Chef's Specials (20 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
S1 EXPFUNC	Let $g(x) = 2^{x+1} - 4$. **(a)** Describe the transformations of the graph of $y = 2^x$. **(b)** Give the horizontal asymptote and the y-intercept. **(c)** Find the x-intercept exactly. **(d)** State the range of g.	**Answer:** (a) left 1, down 4 (b) $y = -4$; $(0, -2)$ (c) $(1, 0)$ (d) $y > -4$ (a) $x + 1$ shifts the graph **left 1**; the -4 shifts it **down 4**. (b) The asymptote moves down with the graph: **$y = -4$**. At $x = 0$: $2^1 - 4 = -2$, so **$(0, -2)$**. (c) $2^{x+1} - 4 = 0$, so $2^{x+1} = 4 = 2^2$, $x + 1 = 2$, and **$x = 1$**. (d) The graph stays above the asymptote: range **$y > -4$**.	2 pts: (a) Left 1 (not right) 2 pts: (a) Down 4 3 pts: (b) Asymptote $y = -4$ 3 pts: (b) y-intercept $(0, -2)$ 3 pts: (c) Sets $g(x) = 0$ and gets $2^{x+1} = 4$ 3 pts: (c) $x = 1$ 4 pts: (d) Range $y > -4$ (not $y > 0$)
S2 LOGDEF	The pH of a solution is $pH = -\log [H^+]$. **(a)** Lemon juice has pH 2. Find $[H^+]$. **(b)** Rainwater has $[H^+] = 2.5 \times 10^{-6}$. Find its pH to the nearest tenth. **(c)** Solution A has pH 3 and solution B has pH 6. How many times greater is $[H^+]$ in A than in B?	**Answer:** (a) 0.01 (b) ≈ 5.6 (c) 1,000 times (a) $\log [H^+] = -2$, so $[H^+] = 10^{-2} = 0.01$ (b) $pH = -\log(2.5 \times 10^{-6}) = -(\log 2.5 - 6) \approx 5.6$ (c) $[H^+] = 10^{-3}$ for A and 10^{-6} for B, and $\frac{10^{-3}}{10^{-6}} = 10^3 = 1{,}000$ times greater.	2 pts: (a) Rewrites as $\log [H^+] = -2$ 3 pts: (a) $[H^+] = 10^{-2} = 0.01$ 3 pts: (b) Substitutes into $-\log(\ldots)$ 3 pts: (b) 5.6 3 pts: (c) Finds both concentrations 3 pts: (c) Divides to get 10^3 3 pts: (c) Answers 1,000 times (A is greater, not B)
S3 LOGPROP	Suppose $\log_b 2 = 0.35$ and $\log_b 3 = 0.55$. Use properties of logs to find **(a)** $\log_b 6$ **(b)** $\log_b 18$ **(c)** $\log_b \frac{2}{9}$. **(d)** Write $2\log_b x + 3\log_b y - \log_b z$ as one logarithm. **(e)** Maya says $\log_b 5 = \log_b 2 + \log_b 3 = 0.90$. Explain her error.	**Answer:** (a) 0.90 (b) 1.45 (c) -0.75 (d) $\log_b \frac{x^2y^3}{z}$ (e) 0.90 is $\log_b 6$, not $\log_b 5$ (a) $\log_b (2 \cdot 3) = 0.35 + 0.55 =$ **0.90** (b) $18 = 2 \cdot 3^2$, so $0.35 + 2(0.55) =$ **1.45** (c) $0.35 - 2(0.55) =$ **-0.75** (d) Power rule, then product and quotient: $\log_b \frac{x^2y^3}{z}$ (e) Adding logs multiplies the inputs: $2 \cdot 3 = 6$, so her sum equals $\log_b 6$. The given values do not tell us $\log_b 5$.	4 pts: (a) Product rule, 0.90 4 pts: (b) Writes $18 = 2 \cdot 3^2$ and uses the power rule: 1.45 4 pts: (c) Quotient rule with $9 = 3^2$: -0.75 4 pts: (d) Powers 2 and 3 in the numerator, z in the denominator 4 pts: (e) Says the sum is log 6 (product rule), not log 5
S4 SOLVE	A drug has 200 mg in the body and a half-life of 6 hours: $A(t) = 200(\frac{1}{2})^{\frac{t}{6}}$. **(a)** How much remains after 18 hours? **(b)** When will 10 mg remain? **(c)** A population $P = 1200e^{0.04t}$ grows continuously. How long to double? Round (b) and (c) to the nearest tenth.	**Answer:** (a) 25 mg (b) ≈ 25.9 hours (c) ≈ 17.3 years (a) 18 hours is 3 half-lives: $200 \cdot \frac{1}{8} =$ **25 mg** (b) $(\frac{1}{2})^{t/6} = \frac{1}{20}$, so $\frac{t}{6} = \frac{\ln 20}{\ln 2}$ and $t = \frac{6 \ln 20}{\ln 2} \approx 25.9$ hours (c) $e^{0.04t} = 2$, so $t = \frac{\ln 2}{0.04} \approx 17.3$ years	4 pts: (a) Sees 3 half-lives: 25 mg 3 pts: (b) Isolates the power: $(\frac{1}{2})^{t/6} = \frac{1}{20}$ 2 pts: (b) Takes a log of both sides 3 pts: (b) 25.9 hours 3 pts: (c) Sets $e^{0.04t} = 2$ 2 pts: (c) Uses ln 3 pts: (c) 17.3 years